The concatenation of transport plans collapses exactly when one kernel is constant on each atom interval

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Editorial note. The counted statement is the author's labelled lem environment at source0.tex lines 1922-1938, original Lemma 4.18; its proof environment opens at line 1940, the line after the blank separator, so the pairing is by adjacency and the retained proof range 1940-1967 is the whole environment.

Editorial note. The retained counter values (section 4, theorem offset 17) make the counted lemma print as Lemma 4.18, immediately after the previously retained Lemma 4.17 at source lines 1907-1912; this article carries no label-number dump, so the number is a counter inference reported with that caveat.

Editorial note. Five environments are retained as context because the statement or the proof refers to them: rem:matrix (source lines 678-680), the definition of the concatenation with its numbered expression (source lines 709-715), rem:finesse (source lines 1465-1489), the notation for the atoms (source lines 1492-1500) and lem:int_fg (source lines 1907-1912).

Editorial note. The retained preamble is the publisher layout adapter followed by the native author macro and theorem block of the pinned source; the article bibliography is deliberately not bundled, since the retained proof carries no citation.

Editorial note. Not counted, withdrawn before assembly succeeded: the retained author proof uses the mathematics macro oldbigcup at source line 1945, which the publisher document class provides and the pinned source never defines. The project rule for the minimal adapter allows plain text abbreviations and pure layout aliases only and requires changing the source when a mathematics macro would have to be supplied, so no adapter line was invented and the card is withdrawn.

Editorial note. r003 note (replaces the r002 note, which had to be withdrawn because it ended up carrying a stray non-printing byte and xelatex refused it with 'Text line contains an invalid character'): r001 failed with 'Undefined control sequence' on the macro oldbigcup (host log, built tex line 253, coming from source0.tex line 1945). That control sequence is defined neither by the author source, nor by the author own pinned preamble (editable-corpus/sources/2088/pinned-required-inputs/author-preamble.tex, whose only RedefinitSymbole calls are for forall and exists), nor by any package installed on the host; a search of the whole installed TeX Live tree found no definition.

The one faithful source is the corpus own already-compiled item for this same problem, editable-corpus/items/2088_author_proof.tex, whose line 7 binds it to the standard big-union operator and whose original compile receipt records ok=true over three passes. The card therefore carries that single line verbatim as its own first preamble span, in a file inside the card directory; the shared source tree is untouched and no macro is invented. Everything else is unchanged from r001.

Editorial note. r004 note: r003 compiled but was rejected by the assembler for two undefined references, eq:concatenation3 and defi:quantile_process, both cited by retained author text and both outside the package. r004 closes the gap with the author own text instead of navigation stubs: the context span that carried the concatenation definition is extended to source line 725 so that it also carries the remark whose displayed formula is labelled eq:concatenation3 (source line 719), and the Proposition/Definition that introduces the quantile measure (source line 1140) is retained as a one-line context span. Because two more theorem-class environments now precede the counted lemma, the initial theorem counter is lowered to 11 so that the counted lemma still prints as the article Lemma 4.18 and lem:int_fg immediately before it as 4.17; this is checked against the job .aux after assembly. No stub is declared for either label, since both are now defined inside the package.

Editorial note. r005 note: r004 was rejected by the assembler with "assembly differs from reviewed context locators" because the six context locators had been written only into assembly.context_spans, while the host validates the card TOP-LEVEL role span lists against assembly.body_spans (prepared_delivery, validate_task: spans == [normalized_span(s) for s in task[role + "_spans"]]). r005 writes the same six locators to the top level, to assembly.context_spans and to the context entries of assembly.body_spans; nothing else changes, and the source locators of the counted statement and proof are untouched.

Editorial note. r006 note: two corrections to r005. The context span for defi:quantile_process was the single line 1140, which opened the prodefinition environment without closing it, so xelatex reported '\begin{prodefinition} on input line 202 ended by \end{document}'; the span now covers lines 1140-1141, the whole environment. And the retained context carries seven theorem-class environments before the counted lemma, not six, so the initial theorem counter is 10 instead of 11: the previous run put lem:int_fg at 4.18 and the counted lemma at 4.19, whereas the target is 4.17 and 4.18, as in the article. Statement, proof and every other locator are unchanged.

Editorial scope. One original result of the pinned publisher article, counted once: the characterisation of the collapse of the concatenation of transport plans by constancy of a kernel on each atom interval (author label lem:distributivite).

Allowed standard prerequisites (editorial).

- Transport plans and their concatenation. A transport plan is a measure on a product with prescribed marginals, and the concatenation of two plans is defined by a disintegration whose expression is numbered eq:concatenation3.. It is the operation whose behaviour is analysed.
- Atoms and the intervals they merge. The atoms of the measure determine intervals of quantile levels on which the quantile function is constant, and

A is their union.. They are the sets on which the constancy condition is imposed.

- Integral inequality for increasing functions. For positive bounded increasing functions the mean of the product dominates the product of the means, with equality only when one factor is constant.. It converts the sign of each summand into the constancy condition.

Remark 4.11. We recall the usual interpretation of the composition as matrix multiplication. If $E = \{x_i\}_{i=1}^{n}$, $E' = \{y_j\}_{j=1}^{n'}$, $E'' = \{z_k\}_{k=1}^{n''}$ are finite, a measure $\theta \in \mathcal{M}(E)$ is a row vector $(\theta(\{x_i\}))_{i=1}^{n} = (\theta_i)_{i=1}^{n}$, a kernel k from E to E' is a matrix $k = ((k_{i,j})_{i=1}^{n})_{j=1}^{n'}$, where $(k_{i,j})_{j=1}^{n'}$ is the measure $k(x_i, \cdot) \in \mathcal{M}(E')$, viewed as a vector, and a function f from E'' to $\mathbb{R}$ is a (column) vector $f = (f(z_j))_{j=1}^{n''}$. Then, taking $k' = ((k'_{j,k})_{j=1}^{n'})_{k=1}^{n''}$ a kernel from E' to E'', $\theta \,.\, k$, $k \,.\, k'$ and $k' \,.\, f$ introduced above have the same sense as products of matrices.

Definition 4.12. If $\mu_i \in \mathcal{P}(E_i)$ for $i \in \{1, 2, 3\}$, if $P_{1,2} \in \mathrm{Marg}(\mu_1, \mu_2)$ and $P_{2,3} \in \mathrm{Marg}(\mu_2, \mu_3)$, their concatenation $P_{1,2} \circ P_{2,3}$ is the unique $R \in \mathcal{P}(\mathbb{R}^3)$ such that for every $(B_1, B_2, B_3) \in \mathcal{B}(E_1) \times \mathcal{B}(E_2) \times \mathcal{B}(E_3)$:

$$R(B_1 \times B_2 \times B_3) = \int_{x \in B_1} \int_{y \in B_2} \int_{z \in B_3} \mathrm{d}\mu_1(x) k_{1,2}(x, \mathrm{d}y) k_{2,3}(y, \mathrm{d}z). \quad (1)$$

In particular, $R \in \mathrm{Marg}((\mu_1, \mu_2, \mu_3)$, $(\mathrm{proj}^{1,2})_{\#}R = P_{1,2}$, and $(\mathrm{proj}^{2,3})_{\#}R = P_{2,3}$.

Remark 4.13. Let $k_{i,j}$ be a disintegration kernel for $P_{i,j}$ and $P_{2,1} := {}^{\mathrm{t}}P_{1,2}$. The right side of (1) also reads: $\iint_{B_1 \times B_2} \int_{B_3} k(y, \mathrm{d}z) \mathrm{d}P_{1,2}(x, y)$, hence also:

$$\int_{y \in B_2} \iint_{x \in B_1,\, z \in B_3} k_{2,1}(y, \mathrm{d}x) k_{2,3}(y, \mathrm{d}z) \mu_2(\mathrm{d}y). \quad (2)$$

Concatenation is "reversed when time is reversed" in the sense that ${}^{\mathrm{t}}P_{2,3} \circ {}^{\mathrm{t}}P_{1,2} = {}^{\mathrm{t}}(P_{1,2} \circ P_{2,3})$; this is immediate after (2). Formula (1) gives immediately that $\circ$ is associative, leading to its following generalization:

$$P_{1,2} \circ \ldots \circ P_{n-1,n}\Big(\prod_{i=1}^{n} B_i\Big) = \int_{(x_i)_i \in \prod_i B_i} \mathrm{d}\mu_1(x_1) k_{1,2}(x_1, \mathrm{d}x_2) \ldots k_{n-1,n}(x_{n-1}, \mathrm{d}x_n).$$

Proposition/Definition 4.14. *If $(\mu_\tau)_{\tau \in \mathcal{T}}$ is a family of measures, there is a unique measure $\mathfrak{Q} \in \mathrm{Marg}((\mu_\tau)_{\tau \in \mathcal{T}})$ such that for every $\tau \neq \sigma$, the transport plan $\mathfrak{Q}^{\tau,\sigma} = (\mathrm{proj}^{\{\tau,\sigma\}})_{\#}\mathfrak{Q}$ is the quantile coupling $\mathfrak{Q}(\mu_\tau, \mu_\sigma)$. We call it the* quantile measure *of $(\mu_\tau)_{\tau \in \mathcal{T}}$.*

Remark 4.15. a. Notice that ${}^{\mathrm{t}}q_s \,.\, q_t = \mathrm{Joint}(\mu_s, {}^{\mathrm{t}}k_s \,.\, k_t) = \mathfrak{Q}^{s,t}$, where $\mathfrak{Q}^{s,t} = \mathfrak{Q}(\mu_s, \mu_t)$, see Definition 4.14. Indeed, ${}^{\mathrm{t}}q_s \,.\, q_t = {}^{\mathrm{t}}q_s \,.\, \mathrm{Id}_{\lambda,2} \,.\, q_t$, then apply (b) below, which will also be useful farther.

b. If some ordered pair (U, V) of variables has law $T \in \mathrm{Marg}(\lambda, \lambda)$, then

$$\mathrm{Marg}(\mu_s, \mu_t) \ni {}^{\mathrm{t}}q_s \,.\, T \,.\, q_t = \mathrm{Law}(G_s(U), G_t(V)) = (G_s \otimes G_t)_{\#}T.$$

Indeed, ${}^{\mathrm{t}}q_s.T.q_t = \mathrm{proj}^{1,4}_{\#}({}^{\mathrm{t}}q_s \circ T \circ q_t)$ and the 4-times process $(G_{\mu_s}(U), U, V, G_{\mu_t}(V))$ has law ${}^{\mathrm{t}}q_s \circ T \circ q_t$ and is Markov, since the σ-fields spanned by U and $\{U, G_{\mu_s}(U)\}$ are the same.

c. From (a) we get ${}^{\mathrm{t}}q_r.q_r = \mathrm{Id}_{\mu_r,2}$, so that ${}^{\mathrm{t}}q_r.q_r.{}^{\mathrm{t}}q_r = {}^{\mathrm{t}}q_r$ and $q_r.{}^{\mathrm{t}}q_r.q_r = q_r$. *However*, $q_r\,.\,{}^{\mathrm{t}}q_r \neq \mathrm{Id}_{\lambda,2}$. Indeed, $k_r\,.\,{}^{\mathrm{t}}k_r$ maps any quantile level $\alpha \in]0,1[$ on itself *except when* $G_r(\alpha)$ is an atom of μ_r. Actually, μ_r-almost surely:

$$k_r\,.\,{}^{\mathrm{t}}k_r(\alpha) = \begin{cases} \delta_\alpha & \text{if } \mu_r(G_r(\alpha)) = 0 \\ (\alpha^+ - \alpha^-)^{-1}\lambda|_{]\alpha^-,\alpha^+[} & \text{if } \alpha \in]\alpha^-,\alpha^+[, \end{cases}$$

where $]\alpha^-,\alpha^+[$ denotes any set $A_{r,x}$ as follows.

Notation 4.16 (Atomic levels, ℓ_r)**.** a. We denote by $A_{r,x}$ the interval $]F_r(x^-),F_r(x)[$ of quantile levels merged by G_r on some atom x of μ_r, and by A_r the set $\bigcup_{\mu_r(x)>0} A_{r,x} \subset [0,1]$ of "atomic levels" of μ_r.

b. We denote $k_r\,.\,{}^{\mathrm{t}}k_r$ described in Remark 4.15(c) by ℓ_r.

Lemma 4.17. *Take $a<b$ in $\mathbb{R}$ and f and g two positive bounded increasing functions from $]a,b[$ to $\mathbb{R}$. Then*

$$\frac{1}{b-a}\int_a^b fg\,\mathrm{d}\lambda \geq \left(\frac{1}{b-a}\int_a^b f\mathrm{d}\lambda\right)\left(\frac{1}{b-a}\int_a^b g\mathrm{d}\lambda\right),$$

with equality if and only if f or g is constant.

Lemma 4.18. *Take $\mu\in\mathcal{P}(\mathbb{R})$, denote by $g=G_\mu$ its quantile function and $F=F_\mu$ its cumulative distribution function. Take $L=\mathfrak{Q}(\lambda,\mu).\mathfrak{Q}(\mu,\lambda)$ and, similarly as in Notation 4.16, let $(b_i)_{i\in\mathcal{I}}$ be the atoms of μ and for each i, A_i be the interval $]F(b_i^-),F(b_i)[$ of quantile levels merged by g on b_i. Finally set $A=\bigcup_{i\in\mathcal{I}} A_i$. Moreover, take*

$$((\nu_i)_i,(\nu'_i)_i) \in \mathcal{P}(\mathbb{R})^{k+k'},$$
$$P \in \mathrm{Marg}(\lambda\lfloor_{[0,1]},\nu_1\otimes\ldots\otimes\nu_k) \quad \textit{and} \quad Q \in \mathrm{Marg}(\lambda_{[0,1]},\nu'_1\otimes\ldots\otimes\nu'_{k'}).$$

Suppose that P and Q have increasing kernel (for $\preceq_{\mathrm{lo}}$ in place of $\preceq_{\mathrm{sto}}$ if $k\geq 2$ or $k'\geq 2$). Then:

a. *${}^{\mathrm{t}}P\,.\,L\,.\,Q$ equals ${}^{\mathrm{t}}P\,.\,Q$ if and only if for each $i\in\mathcal{I}$, at least one of the two kernel functions $x\mapsto k_P(x,\,\cdot\,)$ or $x\mapsto k_Q(x,\,\cdot\,)$ is constant on A_i,*

b. *For any measurable maps $f:\mathbb{R}^k\to\mathbb{R}^k$ and $h:\mathbb{R}^{k'}\to\mathbb{R}^{k'}$,*

$${}^{\mathrm{t}}P\,.\,L\,.\,Q = {}^{\mathrm{t}}P\,.\,Q \implies (f\otimes g\otimes h)_\#({}^{\mathrm{t}}P\circ Q) = (f\otimes g)_\#{}^{\mathrm{t}}P\circ(g\otimes h)_\#Q.$$

Proof. We recall particularly here that we have throughout in mind the analogy between composition of transport plans, or of their kernels, and product of matrices, hinted at in Remark 4.11. In the following, $B_1\subset\mathbb{R}^k$ and $B_3\subset\mathbb{R}^{k'}$ stand for any sets of the type $\prod_j]-\infty,d_j]$ and B_2 for any interval $]-\infty,b]$; (a,b,c) stand for variables in the target of (f,g,h), and (x,y,z) for variables in their sources. The coupling L, equal to ${}^{\mathrm{t}}q_r\,.\,q_r$ (with $\mu=\mu_r$) introduced in Remark 4.15, is described in this remark. It is such that ${}^{\mathrm{t}}P\,.\,L\,.\,Q = ({}^{\mathrm{t}}P\,.\,L)\,.\,(L\,.\,Q) = {}^{\mathrm{t}}((g\otimes\mathrm{Id}_k)_\#P)\,.\,((g\otimes\mathrm{Id}_{k'})_\#Q)$. In turn, for any measurable functions

$f : \mathbb{R}^k \to \mathbb{R}^k$ and $h : \mathbb{R}^{k'} \to \mathbb{R}^{k'}$:

$$\begin{aligned}
&\int_{B_2} k_{(g\otimes f)_\# P}(b, B_1) \,.\, k_{(g\otimes h)_\# Q}(b, B_3)\,\mathrm{d}\mu(b) \\
&\quad= \sum_{i\in\mathcal{I},\, b_i\in B_2} \mu(b_i)[k_{(g\otimes f)_\# P}(b_i, B_1) \,.\, k_{(g\otimes h)_\# Q}(b_i, B_3)] \\
&\qquad\quad + \int_{B_2\setminus\bigcup_i\{b_i\}} k_P(g^{-1}(b), f^{-1}(B_1)) \,.\, k_Q(g^{-1}(b), h^{-1}(B_3))\,\mathrm{d}\mu(b) \\
&\qquad\qquad\qquad\qquad\qquad\qquad \text{as } g \text{ is injective outside of } A \\
&\quad= \sum_{i\in\mathcal{I},\, b_i\in B_2} \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, f^{-1}(B_1))\,\mathrm{d}\lambda(y) \int_{A_i} k_Q(y, h^{-1}(B_3))\,\mathrm{d}\lambda(y) \\
&\qquad\quad + \int_{g^{-1}(B_2)\setminus A} k_P(y, f^{-1}(B_1)) \,.\, k_Q(y, h^{-1}(B_3))\,\mathrm{d}\lambda(y).
\end{aligned} \tag{3}$$

Then, using for instance the expression (2) given in Definition 4.12 of the concatenation, and the fact that for any transport plans R and R', $R \,.\, R' = \mathrm{proj}^{1,3}_\#(R \circ R')$ we get:

$$\begin{aligned}
&({}^{\mathrm{t}}P \,.\, Q - {}^{\mathrm{t}}((g \otimes \mathrm{Id})_\# P) \,.\, (g \otimes \mathrm{Id})_\# Q)(B_1 \times B_3) \\
&\quad= \int_{[0,1]} k_P(y, B_1) k_Q(y, B_3)\mathrm{d}y - \int_{\mathbb{R}} k_{(g\otimes\mathrm{Id})_\# P}(b, B_1) k_{(g\otimes\mathrm{Id})_\# Q}(b, B_3)\mathrm{d}\mu(b) \\
&\quad= \sum_{i\in\mathcal{I}} \int_{A_i} k_P(y, B_1) k_Q(y, B_3)\mathrm{d}y - \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, B_1)\mathrm{d}y \int_{A_i} k_Q(y, B_3)\mathrm{d}y
\end{aligned}$$

by (3) with $B_2 = \mathbb{R}$, $f = \mathrm{Id}$, $h = \mathrm{Id}$—notice in particular that the two terms of the difference, viewed as integrals on $[0, 1]$, differ possibly only on $A \subset [0, 1]$. Now, after Lemma 4.17 applied on each A_i with $f_i : y \mapsto k_P(y, B_1)$ and $g_i : y \mapsto k_Q(y, B_3)$, which are increasing since P and Q have increasing kernel, each term of the sum is non-negative. Thus, equality holds if and only if each vanishes, which again by Lemma 4.17 means that for each i, f_i or g_i is constant. This, holding for any B_1 and B_3, means point (a). For point (b):

$$\begin{aligned}
&{}^{\mathrm{t}}((g \otimes f)_\# P) \circ ((g \otimes h)_\# Q)(B_1 \times B_2 \times B_3) \\
&\qquad= \int_{B_2} k_{(f\otimes g)_\# P}(b, B_1) k_{(g\otimes h)_\# Q}(b, B_3)\,\mathrm{d}\mu(b) \quad \text{by (2)} \\
&\qquad= \int_{g^{-1}(B_2)\setminus A} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y \\
&\qquad\quad + \sum_{i\in\mathcal{I},\, b_i\in B_2} \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, f^{-1}(B_1))\mathrm{d}y \int_{A_i} k_Q(y, h^{-1}(B_3))\mathrm{d}y \quad \text{by (3)},
\end{aligned}$$

but on each A_i, by (a), $k_P(\,.\,, f^{-1}(B_1))$ or $k_Q(\,.\,, h^{-1}(B_3))$ is a constant function, so the last term may be written:

$$\sum_{i\in\mathcal{I},\, b_i\in B_2} \int_{A_i} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y,$$

and therefore the initial expression equals $\int_{g^{-1}(B_2)} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y$, that is to say $(f \otimes g \otimes h)_\#({}^{\mathrm{t}}P \circ Q)(B_1 \times B_2 \times B_3)$. □